

## Restriction in Doubly-Modal Disjunctions\*

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**Abstract** For a disjunction  $\lceil \phi \text{ or } \psi \rceil$  and an information state  $s$ , the right disjunct  $\psi$  is typically evaluated relative information state updated with the negation of the left disjunct – i.e.,  $s[\neg\phi]$ . This observation is central to recent dynamic and information-sensitive analyses of disjunction. However, in doubly-modal disjunctions (DMDs) – expressions of the form  $\lceil \Delta_1 \phi \text{ or } \Delta_2 \psi \rceil$  where  $\Delta_1, \Delta_2 \in \{\Diamond, \Box\}$ , the second disjunct is often evaluated relative to an information state updated with the negation of the left modal’s *prejacent*, i.e.,  $\neg\phi$ , instead of  $\neg\Delta_1 \phi$ . In this paper, we critically evaluate two extant accounts of DMDs due to [Klinedinst & Rothschild \(2012\)](#) and [Meyer \(2015\)](#). We suggest that, in logically complex DMDs, speakers/hearers restrict the right disjunct by an alternative closest in structural complexity to the left disjunct when forced to do so by relevance considerations; otherwise, they restrict the right disjunct by the full left disjunct. We then sketch a novel derivation of conjunctive readings of DMDs based on EXH with innocent inclusion.

**Keywords:** disjunction; formal alternatives; wide scope free choice; QUD

### 1 Introduction

In disjunctions, modals often receive restricted readings – the domain of the modal in the right disjunct is restricted to worlds where the left disjunct is false. As a result, the instance of Disjunctive Syllogism (DS) in (1) is not valid.<sup>1</sup>

- |     |                                                        |                 |
|-----|--------------------------------------------------------|-----------------|
| (1) | John is away at a conference, or he must be on campus. | $p \vee \Box q$ |
|     | John might not be on campus.                           | $\neg \Box q$   |
|     | $\nRightarrow$ John is away at a conference.           | $p$             |

These restricted readings are often used to motivate a semantics for disjunction where the right disjunct is relative to an information state updated with the negation of the left disjunct. For example, in [Klinedinst & Rothschild \(2012\)](#) (K&R henceforth) propose a semantics where disjunction  $\lceil \phi \text{ or } \psi \rceil$  is true at a world  $w$  and information state  $s$  only if the right disjunct  $\psi$  is true at  $w$  and the restricted information state  $s[\neg\phi] = s \cap [\neg\phi]$ .

\* We would like to thank Omar Agha, Fabrizio Cariani, Aron Hirsch, Chenhao Lu, Jessica Mendes, Paolo Santorio, Alexander Williams, and the participants of SALT.

<sup>1</sup> This observation is discussed by both [Klinedinst & Rothschild \(2012\)](#) and [Dorr & Hawthorne \(2013\)](#).

$$(2) \quad \llbracket \phi \text{ or } \psi \rrbracket^{w,s} = 1 \Leftrightarrow \llbracket \phi \rrbracket^{w,s} = 1 \vee \llbracket \psi \rrbracket^{w,s[\neg\phi]} = 1$$

Following Yalcin (2007), K&R assume that non-modal sentences are not sensitive to the information state, and that epistemic modals have a domain semantics, meaning that they quantify over the information state,<sup>2</sup>

$$(3) \quad \llbracket \text{must } \phi \rrbracket^{w,s} = 1 \Leftrightarrow \forall w' \in s : \llbracket \phi \rrbracket^{w',s} = 1$$

Combining these two assumptions, we predict that, in a sentence  $\lceil \phi \text{ or must } \psi \rceil$ , the modal quantifies over an information state restricted to the  $\neg\phi$ -worlds.

$$(4) \quad \llbracket \phi \text{ or must } \psi \rrbracket^{w,s} = 1 \Leftrightarrow \llbracket \phi \rrbracket^{w,s} = 1 \vee \forall w' \in s[\neg\phi] : \llbracket \psi \rrbracket^{w',s} = 1$$

Given this entry, it's easy to see that the instance of DS in (1) fails.

### 1.1 The prejacent problem

However, doubly-modal disjunctions (DMDs) like (5) pose a problem for Klinedinst and Rothschild's proposal; in DMDs, we intuitively want the *prejacent* of the modal in the left disjunct to restrict the domain of the modal in the right disjunct.<sup>3</sup>

- (5)      John must have failed his exam, or he would be happy.     $\Box\phi \text{ or WOLL } \psi$   
           $\rightsquigarrow$  If John passed his exam, he would be happy.                     $\neg\phi \rightarrow \text{WOLL } \psi$   
           $\not\rightsquigarrow$  If John might have passed his exam, he would be happy.  $\Diamond\neg\phi \rightarrow \text{WOLL } \psi$

However, given the entry in (2), Klinedinst and Rothschild's semantics incorrectly predicts that (5) should have the following truth-conditions:

$$(6) \quad \llbracket \Box\phi \text{ or WOLL } \psi \rrbracket^{w,s} = 1 \Leftrightarrow \llbracket \Box\phi \rrbracket^{w,s} = 1 \vee \llbracket \text{WOLL } \psi \rrbracket^{w,s[\Diamond\neg\phi]} = 1$$

Given the meaning in (6), we merely have to learn that it's possible John passed to infer that he's happy. This is absurd. We will call this the *prejacent problem* – in DMDs, we often want the second disjunct to be evaluated relative to an information state updated with the prejacent of the modal in the first disjunct rather than the entire expression.<sup>4</sup>

<sup>2</sup> A similar approach is the bounded semantics proposed by Mandelkern (2019, 2024). Mandelkern's theory closely parallels K&R's but instead assumes classical semantics for modals with definedness conditions concerning restriction from the information state (the local context).

<sup>3</sup> Following Abusch (1985, 1997), we assume *will* and *would* share a common modal morpheme called WOLL which combines with tense to give the meanings of *will* and *would*.

<sup>4</sup> As Yalcin (2007) notes, there are several different ways to make sense of the update effect of epistemic modals in domain semantics. Some of these exacerbate the prejacent problem. For example, if an assertion of  $\Diamond\phi$  simply checks whether the context set is compatible with  $\phi$ , then an assertion either returns the original context  $s[\Diamond\phi] = s$  or the absurd context  $s[\Diamond\phi] = \emptyset$ . This would mean that, not only is the right modal in a DMD not restricted by the prejacent, it's only ever trivially restricted.

The prejacent problem is not a problem unique to K&R’s semantics – a textbook dynamic semantics faces the same problem. For example, in Veltman’s (1996) update semantics, updating an input context  $c$  with  $\lceil \text{might } \phi \rceil$  tests whether  $\phi$  is consistent with  $c$ : returning  $c$  if  $\phi$  is consistent with  $c$ , and the absurd context otherwise.

$$c[\text{might } \phi] := \{w \in c \mid c[\phi] \neq \emptyset\}$$

Given the duality of *might* and *must*, the following entry for epistemic *must* follows from Veltman’s proposal for *might*:<sup>5</sup>

$$c[\text{must } \phi] := \{w \in c \mid c[\phi] = c\}$$

In words,  $\lceil \text{must } \phi \rceil$  tests whether  $c$  accepts  $\phi$ : returning  $c$  if it does. We can then derive the failure of DS in (1) by combining this test-based semantics for *might* with the following entry for disjunction (Beaver 2001; Schlenker 2009):<sup>6</sup>

$$c[\phi \text{ or } \psi] := c[\phi] \cup c[\neg\phi][\psi]$$

However, while these assumptions allow us to predict the invalidity of (1), they yield the wrong prediction concerning DMDs:

$$(7) \quad c[\text{must } \phi \text{ or WOLL } \psi] = c[\text{must } \phi] \cup c[\text{might } \neg\phi][\text{WOLL } \phi]$$

The prejacent problem, moreover, isn’t unique to epistemic modality – the same problem also arises with root modals. For example, given the context in (8), (8a) entails (8b) but does not entail (8c).

- (8) **Context:** John has an A in his philosophy class. He has a final paper due at the end of the term. If John doesn’t write the final paper, he will get a C.
- a. John has to write a paper, or he will get a C.<sup>7</sup>
  - b.  $\rightsquigarrow$  If John doesn’t write a paper, he will get a C.
  - c.  $\nrightarrow$  If John doesn’t have to write a paper, he will get a C.

Given the context, (8c) is clearly false – if writing the paper was not required, then John would receive his current grade, an A. If (8c) were the correct paraphrase of the right disjunct, we’d expect the whole disjunction to be infelicitous.

<sup>5</sup> Veltman (1996) doesn’t discuss epistemic *must*, and never explicitly proposes this entry. For a recent discussion of *must* in update semantics, see Willer (2015).

<sup>6</sup> This entry is independently motivated by facts concerns presupposition filtering in disjunctions.

(i) Either France doesn’t have a king, or the king is in exile.

As Karttunen (1974) first observed, in (i), the existence presupposition of the right disjunct is filtered by the negation of the left disjunct. The prejacent problem causes a problem for standard theories of presupposition filtering too, because they predict that the local context of the right disjunct should be  $\neg\Box\phi$  rather than  $\neg\phi$ .

<sup>7</sup> Some of the speakers we’ve consulted prefer *would* over *will*, but the restriction on the modal is the same in either case.

(9) John must have failed his exam, or he would be happy.  
 $\rightsquigarrow$  John must have failed his exam, and if he hadn't, he would be happy.

Meyer treats the conjunctive interpretation as a scalar inference and sketches an EXH-based derivation – a hypothesis supported by the fact that the conjunctive reading, like other scalar inferences, disappears in downward-entailing environments.<sup>8</sup>

- There are, however, *prima facie* problems with treating the conjunctive reading as a scalar inference. For example, in (11), a context where speakers are explicitly ignorant about John’s goals, the conjunctive reading is blocked.<sup>9</sup>

- While hearers do not interpret (11) as implicating (11b), they do interpret (11) as implicating (11c) where *have to*'s prejacent restricts WOLL.

(12) **Context:** Mary is taking a grad semantics seminar where she has to write one long research paper or two short squibs.

9 Thanks to Paolo Santorio (p.c.) for this example.

- a. Mary has to write a research paper, or she has to write two squibs.
- b.  $\leadsto$  If Mary doesn't write a research paper, she has to write two squibs.
- c.  $\leadsto$  If Mary doesn't write two squibs, she has to write a research paper.

In §4, we suggest that (11) actually does have a conjunctive reading, but it's one where both modals are symmetrically restricted.

### 1.3 Overview

We are not the first to notice the prejacent problem. In §2, we discuss two extant approaches to the problem due to [Klinedinst & Rothschild \(2012\)](#) and [Meyer \(2015\)](#). While both K&R's and Meyer's implementation have different empirical drawbacks, they share a major defect: neither theory is predictive. Neither theory tells us *when* restriction by the prejacent arises. In §3, we propose a novel account of DMDs which combines elements of the accounts proposed by K&R and Meyer. We observe that the availability of prejacent restriction is modulated by the Question Under Discussion (QUD). When the prejacent is at-issue, prejacent restriction is preferred; when the entire disjunct is at-issue, prejacent restriction is dispreferred. We model the variability of restriction as selecting from a set of relevant alternatives to the left disjunct one that differs from it minimally. In §4, we discuss how EXH can derive conjunctive readings for DMDs.

## 2 Previous accounts

There are two extant proposals about how to accommodate DMDs in the literature – one due to [Klinedinst & Rothschild \(2012\)](#), the other due to [Meyer \(2015\)](#). In this section, we highlight some shortcomings of both proposals.

### 2.1 Klinedinst & Rothschild (2012)

K&R briefly acknowledge the prejacent problem. As a response, they propose that the right disjunct can be optionally evaluated relative to an information state updated with a *subclause* of the first disjunct. There are two problems with this simple fix.

**Problem #1: too few options for restriction.** First, as [Cariani \(2017\)](#) points out, K&R's fix doesn't generalize to disjunctions with logically complex disjuncts. Given a sentence of the form  $\lceil (\Diamond\phi \text{ or } \Diamond\psi) \text{ or } \text{WOLL}\chi \rceil$ , Cariani points out that WOLL is often restricted to the  $(\neg\phi \wedge \neg\psi)$ -worlds; for example, in (13), *will* is restricted to worlds where Jones isn't eating lunch and isn't teaching.

- (13) Jones might be eating lunch, or he might be teaching, or he'll be in his office.  
 $\rightsquigarrow$  If Jones isn't eating lunch and isn't teaching, he'll be in his office.

However,  $\lceil \phi \text{ or } \psi \rceil$  is not a sub-clause of  $\lceil \Diamond \phi \text{ or } \Diamond \psi \rceil$ ; so, K&R's simple fix does not predict that the attested reading of (13) is even available.

**Problem #2: not predictive.** Second, K&R offer no principled way for determining when to restrict by a subclause rather than the full clause (as they concede themselves.) This is a problem because, in most contexts, only one of the two restrictions is available. For example, in K&R's example (14), (14b) is seemingly the only possible interpretation the disjunction can receive.

- (14) John must pay alimony, or he will be arrested.  
 a.  $\nrightarrow$  If John isn't required to pay alimony, he will be arrested.  
 b.  $\rightsquigarrow$  If John doesn't pay alimony, he will be arrested.

An example like (14) might be explained by appeal to world knowledge – given our knowledge of laws, we know (14a) is very unlikely to be true, whereas (14b) is very likely to be true. However, most speakers find it's impossible to even access the reading in (14a), suggesting that K&R cannot explain the obligatory nature of preajacent restriction in (14) via post-semantic pragmatic reasoning.

## 2.2 Meyer (2015)

Meyer (2015) proposes that DMDs always contain a subtractive pro-form *else* or its covert counterpart  $\text{pro}_{\text{else}}$ :<sup>10</sup>

- (15) John must have failed his exam, or **else** he would be happy.

Meyer's key idea is that, in sentences like (15), *else* functions as a propositional anaphor, which subtracts the worlds where its antecedent proposition is true from the set of accessible worlds  $R(w)$  for the modal.

$$(16) \quad \llbracket R \text{ else}_k \rrbracket^{g,w} = R(w) - g(k)$$

This set of worlds  $R(w) - g(k)$  then serves as WOLL's domain, leading to the conditional interpretation of (15). To avoid the preajacent problem, Meyer proposes *else* and  $\text{pro}_{\text{else}}$ 's antecedent can be any "salient tree," where a salient tree is defined similarly to the formal alternatives for implicature described by Katzir (2007).

- (17) **SALIENT TREE.** If  $\phi$  can be derived from a given constituent  $\psi$  by deletion or lexical replacement of terminal nodes, then  $\phi$  is salient.

<sup>10</sup> In the nominal domain, it's standard common to treat *else* as a subtractive pro-form to capture how it composes with quantificational DPs (Culicover & Jackendoff 1995).

If we combine these pieces, Meyer’s account predicts that, when  $\phi$  is salient,  $\Delta_2$  is restricted to the accessible  $\neg\phi$ -worlds in  $\lceil \Delta_1\phi \text{ or } \Delta_2\psi \rceil$  where  $\Delta_1, \Delta_2 \in \{\Diamond, \Box\}$ .

Given her syntactic algorithm, Meyer avoids our first critique of K&R; returning to Cariani’s example,  $\lceil \phi \text{ or } \psi \rceil$  can be derived from  $\lceil \Diamond\phi \text{ or } \Diamond\psi \rceil$  by deletion. However, our second objection seems to remain – Meyer’s account isn’t predictive either. In many contexts, restriction by a salient tree isn’t *an* option – it’s the *only* option. However, Meyer’s proposal also faces problems unique to her covert-*else* analysis.<sup>11</sup>

**Problem #1: too many options for restriction.** Whereas K&R’s theory *under-generates* possible readings of DMDs with logically complex disjuncts, Meyer’s theory appears to *overgenerate* possible readings. Let’s focus on a relatively simple example:  $\lceil \Diamond\phi \text{ or } \text{WOLL}\chi \rceil$ . Given Meyer’s Katzir-inspired algorithm, we predict that WOLL in the aforementioned sentence can be restricted by any of the following “salient trees”:  $\Box\phi$ ,  $\Diamond\phi$ , and  $\phi$ . However, restriction by  $\Box\phi$  appears to be unattested; for example, while we can make (18a) and (18b) salient by modulating the QUD, it seems impossible for (18) to ever have the meaning in (18c) (even if the QUD concerns whether taking semantics required.)

- (18) Either John may take semantics, or he’ll have to take logic.
- a.  $\rightsquigarrow$  If John doesn’t take semantics, he’ll have to take logic.
  - b.  $\rightsquigarrow$  If John isn’t permitted to take semantics, he’ll have to take logic.
  - c.  $\nrightarrow$  If John isn’t required to take semantics, he’ll have to take logic.

One might think that this argument highlights an important difference between the alternatives needed to model DMDs and the alternatives needed for implicature calculation. Whereas the alternatives for scalar implicature are those obtainable by deletion and lexical replacement of terminal nodes, DMD restriction is only sensitive to alternatives obtainable by deletion of terminal nodes.<sup>12</sup>

<sup>11</sup> Here’s one potential problem we won’t discuss in-depth: sometimes the inclusion of an overt *else* in a DMD appears to make a semantic difference. For example, Geurts (2005) observes that the DMDs of the form  $\lceil \Box\phi \text{ or } \Diamond\psi \rceil$  are infelicitous:

- (i) #John’s car must be here, or it might be in Mowatt.

However, as Geurts notes, (i) is improved with the addition of an overt *else*.

- (ii) John’s car must be here, or **else** it might be in Mowatt.

Geurts’s contrast is somewhat mysterious given Meyer’s theory: if (i) contains  $\text{pro}_{\text{else}}$ , as Meyer posits, it’s puzzling why it sounds so much worse than its counterpart with an overt *else*.

<sup>12</sup> Sauerland (2004) and Katzir (2007) both highlight several reasons why the alternatives for scalar implicature must be those obtainable both by deletion and lexical replacement. Katzir & Singh (2014) formulate a non-redundancy principle in terms of simplifications.

- (19) An expression  $\phi'$  is a *simplification* of  $\phi$  iff  $\phi'$  can be derived from  $\phi$  by the deletion of terminal nodes.

If we do not consider alternatives derivable from lexical replacement, then  $\Box\phi$  will never be a salient tree for  $\Diamond\phi$ . However, even ignoring the alternatives obtainable by lexical replacement, we still seemingly have too many unattested restrictions. Let's return to Cariani's example:  $\lceil (\Diamond\phi \text{ or } \Diamond\psi) \text{ or } \text{WOLL}\chi \rceil$ . This disjunction's left disjunct has seven simplification alternatives (excluding itself):

$$\Diamond\phi \vee \psi, \phi \vee \Diamond\psi, \phi \vee \psi, \Diamond\phi, \Diamond\psi, \phi, \psi$$

However, many of these simplifications can seemingly never restrict the right disjunct – e.g., (20) never licenses inferences to (20a) or (20b). They're too strong.

- (20) Jones might be eating lunch, or he might be teaching, or he'll be in his office.  
 a.  $\nrightarrow$  If Jones isn't eating lunch, he'll be in his office.  
 b.  $\nrightarrow$  If Jones isn't teaching, he'll be in his office.

And this is not simply because  $\phi \vee \psi$  is weaker than its disjuncts. A conjunctive left disjunct  $\phi \wedge \psi$  leads to the same problem: while each conjunct is weaker than the conjunction, we can't restrict the modal by just one conjunct.

So, while the idea that DMDs are sensitive to formal alternatives is initially plausible, it seems to lead to a proliferation of unattested readings.<sup>13</sup>

**Problem #2: symmetric restriction.** Meyer's proposal hardwires asymmetry into the meaning of DMDs. However, there's good reason to think that symmetric restriction is also possible – for example, (21a) gives rise to symmetric restriction.

- (21) **Context:** Mary is taking a grad semantics seminar where she has to write one long research paper or two short squibs.  
 a. Mary has to write a research paper, or she has to write two squibs.  
 b.  $\rightsquigarrow$  If Mary doesn't write a research paper, she has to write two squibs.  
 c.  $\rightsquigarrow$  If Mary doesn't write two squibs, she has to write a research paper.

However, since *else* is only grammatical in a right disjunct, it's unclear how Meyer's theory can accommodate cases like (21a). In contrast, we can easily amend K&R's semantics to make it symmetric:

<sup>13</sup> We could employ “pruning” of the kind often posited for the formal alternatives relevant to scalar implicature and which we'll employ later (Fox & Katzir 2011; Crnič, Chemla & Fox 2015). However, since pruning is attributed to contextual factors, we would expect that, in some contexts, (20) would entail (20a) and (20b); yet, we don't find such contexts.

$$(22) \quad \llbracket \phi \text{ or } \psi \rrbracket^{w,s} = 1 \Leftrightarrow \llbracket \phi \rrbracket^{w,s[\neg\psi]} = 1 \vee \llbracket \psi \rrbracket^{w,s[\neg\phi]} = 1$$

Examples, like (21a), still pose a problem for K&R because we want symmetric restriction by the prejacent – not symmetric restriction by modals.

### 2.3 Taking stock

In this section, we highlighted some drawbacks of the two extant approaches to DMDs. In particular, K&R’s approach undergenerates possible readings of DMDs, while Meyer’s approach overgenerates possible readings. We also argued that, unlike K&R’s semantics, Meyer’s covert *else* analysis cannot be straightforwardly extended to account for symmetric restriction. However, these accounts both share a major flaw: they’re not predictive. Neither account tells us when the right disjunct should be restricted by a part of the left disjunct rather than the entire left disjunct.

### 3 An alternative-sensitive disjunction

Our positive proposal combines a K&R-style semantics for *or* with a Meyer-style sensitivity to syntactically-derived alternatives. Following K&R, we postulate an information parameter and assume that all modals are sensitive to it by having their domains restricted by it; in contrast, non-modal sentences are not sensitive to this parameter. Following Meyer, we appeal to simplifications of disjuncts.

The key question is, once we get to a set of simplifications, how do we determine which simplification gets picked as the restrictor? The innovation of our proposal is that the restrictor is determined in two steps: we first constrain the set of simplification alternatives by relevance, and then find, among the relevant simplifications, the structurally closest one to the original disjunct.<sup>14</sup>

More concretely, we can capture this idea using a *selection function*  $\text{sel} : \mathcal{P}(W) \times \mathcal{P}(\mathcal{P}(W)) \rightarrow \mathcal{P}(W)$ :

$$(23) \quad \llbracket \phi \text{ or } \psi \rrbracket^{w,s} = \llbracket \phi \rrbracket^{w,s} = 1 \vee \llbracket \psi \rrbracket^{w,s[\neg\text{sel}(\phi, \text{alt}_R(\phi))]} = 1$$

where  $\text{alt}_R(\phi)$  is the set of simplification alternatives of  $\phi$  (including  $\phi$ ), ordered by structural complexity of the corresponding syntactic objects (Katzir 2007), and restricted to a set of contextually relevant propositions  $R$ :

$$\text{alt}_R(\phi) := R \cap \mathcal{S}(\phi)$$

Given  $\phi$  and a set  $\text{alt}_R(\phi)$ , the selection function  $\text{sel}$  outputs the member of the latter (**Inclusion**) closest in structural complexity to  $\phi$ . Intuitively, that’s the relevant

<sup>14</sup> Strictly speaking, relevant simplification alternatives are not syntactic objects but propositions, but we assume that the syntactic objects (the simplifications) they are interpreted from are ordered by complexity and that the propositions can inherit the ordering.

simplification that involves the least steps of deletion based on  $\phi$ . If the default restriction  $\phi$  is relevant, it will be selected (**Centering**) (Stalnaker 1968). For nothing is structurally closer to it than itself. If  $\phi$  is not relevant, the structurally closest relevant simplification will be selected.

- **Inclusion:**  $\text{sel}(\phi, \text{alt}_R(\phi)) \in \text{alt}_R(\phi)$
- **Centering:** If  $\phi \in \text{alt}_R(\phi)$ ,  $\text{sel}(\phi, \text{alt}_R(\phi)) = \phi$ .

It's easy to slightly tweak (23) to allow for the possibility of symmetric restriction:

$$(24) \quad \llbracket \phi \text{ or } \psi \rrbracket^{w,s} = \llbracket \phi \rrbracket^{w,s[\neg \text{sel}(\psi, \text{alt}_R(\neg \psi))]} = 1 \vee \llbracket \psi \rrbracket^{w,s[\neg \text{sel}(\phi, \text{alt}_R(\phi))]} = 1$$

Note that, given non-modal disjuncts whose truth-conditions are not sensitive to  $s$ , both (23) and (24) are equivalent to Boolean disjunction.<sup>15</sup>

Given a DMD  $\ulcorner \Delta_1 \phi \text{ or } \Delta_2 \psi \urcorner$  where  $\Delta_1, \Delta_2 \in \{\Diamond, \Box\}$ ,  $\Delta_2$ , the selection function-based semantics in (23) predicts that restriction by  $\neg \phi$  is only licensed when  $\Delta_1 \phi \notin R$  – i.e., when  $\Delta_1 \phi$  is not relevant.<sup>16</sup> As we'll show in §3.1, restriction is modulated by the Question Under Discussion (QUD) (Roberts 2012), which provides a natural notion of relevance. In particular, when only the prejacent is at-issue, restriction by the prejacent is preferred, while restriction by the entire modalized claim is preferred when the modalized claim itself is at-issue. We then motivate our use of selection function to obtain a predictive theory of DMDs.

### 3.1 The QUD modulates prejacent restriction

The availability of prejacent restriction appears subject to the following constraint:

**At-Issue-ness Generalization.** Given a DMD of the form  $\ulcorner \Delta_1 \phi \text{ or } \Delta_2 \psi \urcorner$  where  $\Delta_1, \Delta_2 \in \{\Diamond, \Box\}$ ,  $\Delta_2$  is restricted by  $\neg \phi$  when  $\phi$  is at-issue and restricted by  $\neg \Delta_1 \phi$  when  $\Delta_1 \phi$  is at-issue.

There's compelling empirical evidence for this generalization. First, it accounts for why in our examples involving epistemics, prejacent restriction almost invariably arises. It is plausible that epistemic modals typically answer questions concerning their prejacent. For instance, the following question-answer pair is fairly natural.

- (25) a. Q: Did John fail his exam?  
 b. A: He must have failed.

<sup>15</sup> We'll make a stipulation that, when  $\text{alt}_R(\phi) = \emptyset$ :  $\text{sel}$  outputs  $\perp$ , which gets negated in  $s[\neg \text{sel}(\phi, \text{alt}_R(\phi))]$  so that  $s$  is effectively updated by a tautology.

<sup>16</sup> This is, again, only true if we assume **Inclusion**.

In contrast, questions concerning the status of epistemic modal claims, though possible, are far less natural – e.g., a corpus study by [Hacquard & Wellwood \(2012\)](#) found that, in matrix questions, epistemic *must* only occurred 34 times, while root *must* occurred 243 times.<sup>17</sup>

Second, there is variability when root modals are involved, correlating with the QUD. In a context like (26), where the QUD concerns whether or not the requirement in the left disjunct holds, restriction by the entire modal claim is preferred.

- (26) **Context:** John’s grades are currently in the A-range, but if he needs to write a paper for the class, John’s grades will drop to B+. However, the professor is contemplating dropping the paper requirement for everyone.
- a. (Either) John has to write a paper, or he will get an A.
  - b.  $\rightsquigarrow$  If John doesn’t have to write a paper, he will get an A.
  - c.  $\nrightarrow$  If John doesn’t write a paper, he will get an A.

In comparison, in a context like (27), where the QUD concerns whether the prejacent of the modal in the left disjunct is true (e.g. *Will John write a paper?*), restriction by the prejacent is preferred.

- (27) **Context:** John’s grades are currently in the A-range, but he needs to write a final paper for the class. Otherwise, his grades will drop to a B+. However, John is contemplating blowing off the final paper to party with his friends.
- a. (Either) John has to write a paper, or he will get a B+.
  - b.  $\nrightarrow$  If John doesn’t have to write a paper, he will get a B+.
  - c.  $\rightsquigarrow$  If John doesn’t write a paper, he will get a B+.

Hence, we see that, when the prejacent is conversationally relevant, restriction with the prejacent is obligatory.

### 3.2 The At-Issue Constraint on restriction

In particular, given a disjunction  $\lceil \Box\phi \text{ or } \text{WOLL } \psi \rceil$ , prejacent restriction is preferred if the QUD concerns whether  $\phi$  is true, while restriction by the whole modal is preferred if the QUD concerns whether  $\Box\phi$  is true. So, instead of considering all possible simplifications, we only want to consider simplifications that are at-issue:

**At-Issue Constraint.** For a disjunction  $\lceil \phi \text{ or } \psi \rceil$  given QUD  $Q$ : restriction by  $p \in \mathcal{S}(\phi)$  is allowed only if  $p$  is relevant to  $Q$ , i.e.,  $p$  is identical to some union of cells of  $Q$ .

<sup>17</sup> Hacquard and Wellwood also show that *might* (523), which typically receives an epistemic interpretation, is far less prevalent in matrix questions than *can* (17,971), which typically receives a root interpretation. Both contrasts are statistically significant – i.e.,  $p < 0.05$ .

Following Groenendijk & Stokhof (1984), we assume that questions are partitions on the set of worlds  $W$ . We can then define the *relevancy set*  $R^Q$  as follows:

$$(28) \quad R^Q := \{Z \mid \exists X, Y \in Q : X \cup Y = Z \wedge Z \neq W\}$$

In other words,  $R^Q$  contains all the partial and complete answers to  $Q$ . For example, given a question with three answers like (29a),  $R^Q$  consists of all the answers that eliminate one or two cells of  $Q$ .<sup>18</sup>

$$(29) \quad \begin{aligned} \text{a. } Q &= \{A, B, C\} \\ \text{b. } R^Q &= \{A, B, C, A \cup B, A \cup C, B \cup C\} \end{aligned}$$

We can tweak our definition of  $\text{alt}$  so that we only consider the alternative restrictions which are relevant to the QUD  $Q$ :

$$(30) \quad \text{alt}_{R^Q}(\phi) := R^Q \cap \mathcal{S}(\phi)$$

For example, consider  $\lceil \Box\phi$  or  $\text{WOLL } \psi \rceil$  given the QUD is  $Q = \{\phi, \neg\phi\}$ . Given the At-Issue Constraint, we're left with only prejacent:

$$(31) \quad \begin{aligned} \text{a. } \mathcal{S}(\Box\phi) &= \{\phi, \Box\phi\} \\ \text{b. } \text{alt}_{R^Q}(\Box\phi) &= \{\phi\} \end{aligned}$$

Here,  $\text{sel}$  given  $\text{alt}_{R^Q}(\Box\phi)$  will output  $\phi$ . Hence, our theory makes a concrete prediction about when prejacent restriction occurs in DMD: speakers will restrict with the prejacent only when the modalized claim itself is irrelevant to the QUD.

This captures our original example (5) which we repeat in (32)

$$(32) \quad \text{John must have failed his exam, or he would be happy.}$$

If we assume that the salient QUD in (32) is the polar question  $Q = \{\text{fail}, \text{pass}\}$ , then  $\mathcal{S}(\Box\text{fail}) = \{\Box\text{fail}, \text{fail}\}$ , and  $\text{alt}_{R^Q}(\Box\text{fail}) = \{\text{fail}\}$ ; so,  $\text{sel}$  will select *fail*, yielding the following truth-conditions:

$$(33) \quad \llbracket (32) \rrbracket^{w,s} = 1 \Leftrightarrow \llbracket \Box\text{fail} \rrbracket^{w,s} = 1 \vee \llbracket \text{WOLL happy} \rrbracket^{w,s[\neg\text{fail}]} = 1$$

### 3.3 Complex examples

In our examples so far,  $\text{alt}_{R^Q}(\phi)$  only outputs a singleton set. But there are cases where the multiple simplifications are relevant.

To see how the selection function is helpful when multiple simplifications are relevant, consider a disjunction  $\lceil (\phi \text{ or } \psi) \text{ or } \Delta\chi \rceil$  where  $\Delta$  is a modal. Suppose the simplifications in  $\{\phi \vee \psi, \phi, \psi\}$  are all relevant. The empirical picture is that

<sup>18</sup> Note: since we're assuming questions are partitional,  $A \cup B \cup C = W$  – i.e., it's true in all possible worlds; hence,  $A \cup B \cup C \notin R^Q$ .

we don't just restrict with any disjunct individually, but with the whole disjunction. The selection function predicts this: when it's given  $\phi \vee \psi$ , and a set of relevant simplifications, it will look for the simplification that involves minimal deletion—in this case it will just select  $\phi \vee \psi$ .

This extends to our running example of a logically-complex disjunction (13) which we repeat in (34).

- (34) Jones might be eating lunch, or he might be teaching, or he'll be in his office.  
 $\rightsquigarrow$  If Jones isn't eating lunch and isn't teaching, he'll be in his office.

The left disjunct of (34) has the following eight simplifications (including itself):

$$\begin{aligned} &\Diamond lunch \vee \Diamond teaching, lunch \vee \Diamond teaching, \Diamond lunch \vee teaching, \\ &\Diamond lunch, \Diamond teaching, lunch \vee teaching, lunch, teaching \end{aligned}$$

Of these simplifications, we exclude all those that are irrelevant to  $Q$  as in (35a). We're left with the set of relevant alternatives in (35b). Of these,  $\text{sel}$  returns ( $lunch \vee teaching$ ) since it's structurally the closest to the left disjunct ( $\Diamond lunch \vee \Diamond teaching$ ).

- (35) a. What is John doing now?  $Q = \{lunch, teaching, office\}$   
 b.  $\text{alt}_{RQ}(\Diamond lunch \vee \Diamond teaching) = \{lunch \vee teaching, lunch, teaching\}$   
 c.  $\text{sel}(\Diamond lunch \vee \Diamond teaching, \text{alt}_{RQ}(\Diamond lunch \vee \Diamond teaching)) = lunch \vee teaching$

So, when coupled with an independently motivated pragmatic constraint on formal alternatives, our selection function-based avoids the over- and under-generation problems that plague K&R and Meyer.<sup>19</sup>

#### 4 The conjunctive reading via Innocent Inclusion

In previous sections, we said certain cases of DMDs don't give rise to conjunctive readings, on the assumption that disjunctions involve asymmetric restriction. We repeat our example (11) as (36). However, if we adopt the assumption that disjunctions involve symmetric restriction, (36) *does* appear to license a conjunctive reading. In particular, while (36a) doesn't imply (36b), it does seem to imply (36c) where *has to* is restricted by the right disjunct.

- (36) **Context:** John is a very bad guitarist. However, we are ignorant about John's goals – he might just want to play the guitar for fun, or he might want to become a professional guitarist.  
 a. Either John has to practice 6 hours a day, or he won't become a professional.

<sup>19</sup> One can also verify that the account also makes the correct predictions when the left disjunct is a conjunction.

- b.  $\nrightarrow$  John has to practice 6 hours a day.
- c.  $\rightsquigarrow$  To become a professional, John has to practice 6 hours a day.
- d.  $\rightsquigarrow$  If John doesn't practice 6 hours a day, he won't become a professional.

In this section, we show that the conjunctive readings of DMDs can be derived by the exhaustification with innocent inclusion, due to the internal dynamicity of K&R's entries for *and* and *or*. We assume symmetric restriction for *or* to not derive an overly strong reading.

First, in K&R's entries for conjunction (37a) and disjunction with symmetric restriction (37b), when  $\psi$  contains a modal, its restriction is very different in the two cases. In disjunction,  $\psi$  is restricted by  $\neg\phi$ , while it's restricted by  $\phi$  in a conjunction.

$$(37) \quad \begin{aligned} \text{a. } \llbracket \phi \text{ or } \psi \rrbracket^{w,s} = 1 &\Leftrightarrow \llbracket \phi \rrbracket^{w,s[\neg\psi]} = 1 \vee \llbracket \psi \rrbracket^{w,s[\neg\phi]} = 1 \\ \text{b. } \llbracket \phi \text{ and } \psi \rrbracket^{w,s} = 1 &\Leftrightarrow \llbracket \phi \rrbracket^{w,s} = 1 \wedge \llbracket \psi \rrbracket^{w,s[\phi]} = 1 \end{aligned}$$

Hence, when  $\psi$  is a modal claim, the following DeMorgan equivalency can fail:

$$\neg(\phi \wedge \psi) \Leftrightarrow \neg\phi \vee \neg\psi$$

In other words, for a modal  $\psi$ , it may be that either  $\llbracket \phi \rrbracket^{w,s} = 0$  or  $\llbracket \psi \rrbracket^{w,s[\phi]} = 0$  and yet both  $\llbracket \phi \rrbracket^{w,s[\neg\psi]} = 1$  and  $\llbracket \psi \rrbracket^{w,s[\neg\phi]} = 1$  – e.g., when  $\psi$  is true relative to the  $\neg\phi$ -worlds in  $s$  but false relative to the  $\phi$ -worlds in  $s$ .

Given Bar-Lev and Fox's semantics for EXH, the fact that  $\neg(\phi \text{ and } \psi)$  is not equivalent to  $(\neg\phi \text{ or } \neg\psi)$  in DMDs is sufficient to license conjunctive strengthening. Bar-Lev & Fox (2017) argue that EXH quantifies over both innocently excludable (IE) and innocently includable alternatives (II). And IIs are computed based on IEs.

$$(38) \quad \begin{aligned} \llbracket \text{EXH}_C \rrbracket^w &= \lambda p_{\langle s,t \rangle}. p(w) = 1 \wedge \forall q \in IE(p, C) : \neg q(w) \wedge \forall r \in II(p, C) : r(w) \\ \text{a. } IE(p, C) &= \bigcap \{ C' \subseteq C : C' \text{ is a maximal subset of } C, \text{ s.t.} \\ &\quad \{ \neg q : q \in C' \} \cup \{ p \} \text{ is consistent} \} \\ \text{b. } II(p, C) &= \bigcap \{ C'' \subseteq C : C'' \text{ is a maximal subset of } C, \text{ s.t.} \\ &\quad \{ r : r \in C'' \} \cup \{ \neg q : q \in IE(p, C) \} \cup \{ p \} \text{ is consistent} \} \end{aligned}$$

Given this semantics for EXH, a disjunction can only be strengthened to conjunction when its alternative set isn't closed under conjunction. In the absence of conjunctive closure, there may not be an innocently excludable conjunction to block further strengthening via innocent inclusion. For example, given the conjunctively closed alternative set in (39a), EXH applied to *or* will return exclusive disjunction.

$$(39) \quad \begin{aligned} \text{a. } C &= \{ \phi \vee \psi, \phi \wedge \psi, \phi, \psi \} \\ \text{b. } \text{EXH}_C(\phi \text{ or } \psi) &= (\phi \vee \psi) \wedge \neg(\phi \wedge \psi) \end{aligned}$$

However, if  $\phi \wedge \psi$  were missing from  $C$ , as it is in (40a), then EXH applied to *or* would return  $\phi \wedge \psi$  because  $\phi$  and  $\psi$  are both innocently includable.<sup>20</sup>

- (40) a.  $C = \{\phi \vee \psi, \phi, \psi\}$   
 b.  $\text{EXH}_C(\phi \text{ or } \psi) = (\phi \vee \psi) \wedge \phi \wedge \psi = \phi \wedge \psi$

It's important to note that closure under conjunction is essentially a semantic notion pertaining to the effect of innocent exclusion on subsequent inclusion. Only when there is a conjunctive alternative whose negation ensures that the two disjuncts won't both be true, would inclusion of both disjuncts be blocked.

Given K&R's entries in (37), when  $\phi$  or  $\psi$  is a modal claim, the alternative set of  $\lceil \phi \text{ or } \psi \rceil$  is not closed under conjunction. Intuitively, given that  $\lceil \phi \text{ or } \psi \rceil$  asserts  $\llbracket \phi \rrbracket^{w,s[\neg\psi]} \vee \llbracket \psi \rrbracket^{w,s[\neg\phi]}$ , the two disjunct alternatives should mean  $\llbracket \phi \rrbracket^{w,s[\neg\psi]}$  and  $\llbracket \psi \rrbracket^{w,s[\neg\phi]}$  respectively. If these are available alternatives, then the innocent exclusion of  $\lceil \phi \text{ and } \psi \rceil$  won't hinder their inclusion. This is precisely due to the failure of DeMorgan we saw above. Given K&R's entries, denying the *and*-alternative doesn't imply that the two disjunct alternatives aren't both true.

So, given Bar-Lev and Fox's algorithm, K&R's entries for disjunction and conjunction license strengthening when  $\phi$  and  $\psi$  are modal, provided that restricted disjunct alternatives are available. If the alternative set is as in (41a), with the interpretation of each alternative spelled out, we get conjunctive reading. To improve readability, we abbreviate  $\llbracket \phi \rrbracket^{w,s} = 1$  as  $\phi$  and  $\llbracket \phi \rrbracket^{w,s[\psi]} = 1$  as  $\psi \Rightarrow \phi$ .

- (41) a.  $C = \{(\neg\psi \Rightarrow \phi) \vee (\neg\phi \Rightarrow \psi), \phi \wedge (\phi \Rightarrow \psi), \neg\psi \Rightarrow \phi, \neg\phi \Rightarrow \psi\}$   
 b.  $\text{EXH}_C(\phi \text{ or } \psi) = ((\neg\psi \Rightarrow \phi) \vee (\neg\phi \Rightarrow \psi)) \wedge \neg(\phi \wedge (\phi \Rightarrow \psi))$   
 $\quad \quad \quad \wedge (\neg\psi \Rightarrow \phi) \wedge (\neg\phi \Rightarrow \psi)$   
 $\quad \quad \quad = (\neg\psi \Rightarrow \phi) \wedge (\neg\phi \Rightarrow \psi) \wedge \neg(\phi \wedge (\phi \Rightarrow \psi))$   
 $\quad \quad \quad = \llbracket \phi \rrbracket^{w,s[\neg\psi]} \wedge \llbracket \psi \rrbracket^{w,s[\neg\phi]} \wedge \neg(\llbracket \phi \rrbracket^{w,s} \wedge \llbracket \psi \rrbracket^{w,s[\phi]})$

However, when  $\phi$  and  $\psi$  are non-modal, K&R's entries for conjunction and disjunction are equivalent to their Boolean counterparts, and  $\phi$  and  $\psi$  are, accordingly, no longer innocently includable. So, if we assume symmetric restriction, not overly strong conjunctive readings of DMDs can be derived from EXH with innocent inclusion without also predicting conjunctive readings of non-modal disjunctions.

To derive the intended conjunctive reading in (41), we posited that the alternative set contained alternatives interpreted as  $\llbracket \phi \rrbracket^{w,s[\neg\psi]}$  and  $\llbracket \psi \rrbracket^{w,s[\neg\phi]}$ ; however, there is a complication here: these are not simplifications of the original disjunction (i.e., alternatives generated by deletion.) If we only consider simplifications, the left and

<sup>20</sup> There's strong empirical evidence that this occurs in languages without conjunction – e.g., Warlpiri has a single coordinator *manu* which is truth-conditionally equivalent to disjunction, but can receive a conjunctive reading via pragmatic strengthening (Bowler 2014).

right disjuncts will both be evaluated relative to the initial  $s$ , without the desired restrictions on  $s$ . We need to postulate internally dynamic replacement alternatives of *or* – **L**, which asserts the restricted left disjunct, and **R**, which asserts the restricted right disjunct. In fact, this move is independently supported. Free Choice-type inferences are licensed even when the right disjunct contains an anaphora dependent on an antecedent in the left disjunct (Elliott & Sudo 2025). However, simplifications break the anaphoric dependency between the disjuncts. For example, as Buccola & Chemla (2019) point out, deleting the left disjunct of (42a) leaves us with the alternative in (42b).

- (42) a. Every dad<sub>i</sub> called [his<sub>i</sub> daughter]<sub>j</sub> or her<sub>j</sub> dog.  
 b. Every dad<sub>i</sub> called her<sub>j</sub> dog.

However, intuitively, we want the alternatives of (42a) to be “Every dad called his daughter” and “Every dad called his daughter’s dog.”

As far as exhaustification goes, our disjunction behaves exactly the same as K&R’s disjunction. When there is no modal in either disjunct, *or* forms a proper scale with *and* and exhaustification gives rise to an exclusive reading of *or*.<sup>21</sup> Whenever there is a modal disjunct that’s information sensitive, the key DeMorgan equivalency fails due to the internal dynamic differences between *and* and *or*. Hence, we can derive the conjunctive reading of modal disjunctions as follows

- (43) a.  $C = \{\llbracket \phi \text{ or } \psi \rrbracket, \llbracket \phi \text{ and } \psi \rrbracket, \llbracket \phi \text{ L } \psi \rrbracket, \llbracket \phi \text{ R } \psi \rrbracket\}$   
 b.  $\llbracket \phi \text{ L } \psi \rrbracket^{w,s} = 1 \Leftrightarrow \llbracket \phi \rrbracket^{w,s[\neg \text{sel}(\phi, \text{alt}_R(\psi))]} = 1$   
 c.  $\llbracket \phi \text{ R } \psi \rrbracket^{w,s} = 1 \Leftrightarrow \llbracket \psi \rrbracket^{w,s[\neg \text{sel}(\phi, \text{alt}_R(\phi))]} = 1$   
 d.  $\text{EXH}_C(\phi \text{ or } \psi) = \llbracket \phi \rrbracket^{w,s[\neg \text{sel}(\phi, \text{alt}_R(\psi))]} \wedge \llbracket \psi \rrbracket^{w,s[\neg \text{sel}(\phi, \text{alt}_R(\phi))]} \wedge \neg \llbracket \phi \text{ and } \psi \rrbracket^{w,s}$

Hence, for DMDs, if we assume symmetric restriction, we predict conjunctive strengthening in the presence of EXH as desired.

However, there does seem to be a slight overgeneration problem if disjunction triggers obligatory exhaustification (Magri 2009, 2011; Del Pinal 2021). For disjunctions with a single modal, if EXH is obligatory, then we predict that disjunctions with a single modalized disjunct are always conjunctively strengthened. For example, suppose that  $\phi$  is non-modal, but  $\psi$  is modal – since  $\phi$  isn’t sensitive to  $s$ ,  $\psi \Rightarrow \phi$  is equivalent to  $\phi$ .

- (44) a.  $C = \{\phi \vee (\neg \phi \Rightarrow \psi), \phi \wedge (\phi \Rightarrow \psi), \phi, \neg \phi \Rightarrow \psi\}$   
 b.  $\text{EXH}_C(\phi \text{ or } \psi) = \phi \vee (\neg \phi \Rightarrow \psi) \wedge \neg(\phi \wedge (\phi \Rightarrow \psi)) \wedge \phi \wedge (\neg \phi \Rightarrow \psi)$   
 $= \phi \wedge (\neg \phi \Rightarrow \psi) \wedge \neg(\phi \Rightarrow \psi)$

21 For how to ensure this result, see fn.14.

$$= \llbracket \phi \rrbracket^{w,s} \wedge \llbracket \psi \rrbracket^{w,s[\neg\phi]} \wedge \neg \llbracket \psi \rrbracket^{w,s[\phi]}$$

However, in some disjunctions with a single modalized disjunct, conjunctive strengthening appears to be non-obligatory – e.g., while K&R’s example (45a) seems to entail the left disjunct, (45b) doesn’t.

- (45) a. I’m away at a conference, or I would meet with you.  
b. Either John is away at a conference, or he must be home.

The difference between disjunctions like (45a) and (45b) leads K&R to posit that *or* is ambiguous – in what they call “non-tabular” disjunctions, like (45a), they posit that *or* has the conjunctive semantics in (46).

$$(46) \quad \llbracket \phi \text{ or}_{\text{NT}} \psi \rrbracket^{w,s} = 1 \Leftrightarrow \llbracket \phi \rrbracket^{w,s} = 1 \wedge \llbracket \psi \rrbracket^{w,s[\neg\phi]} = 1$$

Of course, if possible, we would like to avoid having to adopt an ambiguity theory like the one K&R suggest. Moreover, postulating  $\text{or}_{\text{NT}}$  blocks conjunctive readings in DMDs. However, if implicatures are generated blindly, the contrast between ordinary disjunctions and “non-tabular” disjunction is puzzling. We would expect disjunctions like (45b) to pattern like their “non-tabular” counterparts. However, (45a) and (45b) notably differ in the flavor of the modal. A natural hypothesis is that, in (45a) the counterfactual morphology of *would*, what von Fintel & Iatridou (2023) call X-marking, doesn’t trigger a presupposition that conflicts with the first disjunct, whereas in (45b) the indicative morphology of *must* triggers a presupposition that conflicts with the first disjunct.<sup>22</sup>

For these reasons, we remain agnostic about whether the relevant implicature should be treated as obligatory. We also remain agnostic about whether conjunctive strengthening should be derived from EXH with innocent inclusion. Our goal in this section was merely to sketch *a way* to derive the conjunctive reading of DMDs which avoids the objection we leveled against Meyer.

## 5 Conclusion

This paper started with the observation that doubly-modal disjunctions (DMDs) pose a non-trivial compositional problem for dynamic and informational theories of disjunction. In §2, we surveyed two approaches to the problem – one due to Klinedinst & Rothschild (2012) allows any sub-clause of the left disjunct to restrict the right disjunct’s information state, the other due to Meyer (2015) posits that any formal alternative to the left disjunct can restrict the right disjunct’s information state.

<sup>22</sup> Following Stalnaker (1975), it’s commonly assumed that indicative morphology triggers the presupposition that the modal base is a subset of the context set – the worlds compatible with the propositions that are common ground in the conversation.

While K&R’s theory undergenerates possible readings of DMDs, Meyer’s theory overgenerates possible readings. Moreover, neither theory is predictive – neither theory tells us why preadjacent restriction is sometimes obligatory in DMDs, while restriction by the full disjunct is obligatory other times. In §3, we proposed a novel solution to the preadjacent problem. In particular, we argued that, in logically-complex DMDs, speakers/hearers restrict the right disjunct by an alternative closest in structural complexity to the left disjunct when forced to do so by context; otherwise, they restrict the right disjunct with the full left disjunct. In §4, we sketched a way to derive the conjunctive reading of DMDs based on EXH with innocent inclusion which exploits the fact that K&R’s semantics for conjunction and disjunction are internally dynamic.

While we’ve focused exclusively on disjunctions of modals, [McHugh \(2024\)](#) observes that the same phenomenon occurs in disjunctions of conditionals – e.g., in (47a) the conditional in the right disjunct is restricted by the negation of the left antecedent, while in (47b) the conditional in the right disjunct is restricted by the negation of the consequent of the conditional in the left disjunct.

- (47) a. If there’s no bathroom in this house, I’ll go home, or if it’s in a funny place, I’ll ask the host for directions.  
 b. If we hire architect A, our new office will have no bathroom, or if we hire architect B, it will be in a strange place.

Just as DMDs license conjunctive strengthening, disjunctions of conditionals with the same consequent also license conjunctive strengthening – i.e., Simplification of Disjunctive Antecedents (SDA) where *or* takes wide-scope ([Geurts 2005](#)).<sup>23</sup>

**Wide-Scope SDA.**  $(\phi > \chi) \vee (\psi > \chi) \models (\phi > \chi) \wedge (\psi > \chi)$

- (48) If a heart had been drawn, the card would have been red; or, if a diamond had been drawn, the card would have been red.  
 a.  $\rightsquigarrow$  If a heart had been drawn, the card would have been red.  
 b.  $\rightsquigarrow$  If a diamond had been drawn, the card would have been red.

For reasons of space, we won’t show how to extend our theory of DMDs to disjunctions of conditionals, but we believe that our account can accommodate both McHugh’s data and derive Wide-Scope SDA.

<sup>23</sup> Note that, like the conjunctive reading of DMDs, Wide-Scope SDA doesn’t appear to hold in downward-entailing contexts.

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